

ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)

Assignment 3 – Answer Key

Course: Mathematics in India (IKS 4001) | Semester VIII | B.Tech | AY 2025-2026

Topic: Ganitapada, Mathematics moves to South, Nila Mathematics

Q1. Using Kuttaka method (Euclidean Algorithm) find a solution to the Diophantine equation: $-13x + 60y = 1$

The Kuttaka ('pulveriser') method, attributed to Aryabhata, solves linear Diophantine equations $ax + by = c$ using a systematic process equivalent to the Extended Euclidean Algorithm.

Step 1: Apply the Euclidean Algorithm to find $\gcd(60, 13)$:

$$60 = 4 \times 13 + 8$$

$$13 = 1 \times 8 + 5$$

$$8 = 1 \times 5 + 3$$

$$5 = 1 \times 3 + 2$$

$$3 = 1 \times 2 + 1$$

$$2 = 2 \times 1 + 0 \rightarrow \gcd(60, 13) = 1 \checkmark \text{ (solution exists)}$$

Step 2: Back-substitute (Extended Euclidean) to express 1 as a combination of 60 and 13:

$$1 = 3 - 1 \times 2$$

$$= 3 - 1 \times (5 - 3) = 2 \times 3 - 5$$

$$= 2 \times (8 - 5) - 5 = 2 \times 8 - 3 \times 5$$

$$= 2 \times 8 - 3 \times (13 - 8) = 5 \times 8 - 3 \times 13$$

$$= 5 \times (60 - 4 \times 13) - 3 \times 13 = 5 \times 60 - 23 \times 13$$

Step 3: Match with $-13x + 60y = 1$:

$$\blacksquare 60(5) + 13(-23) = 1 \rightarrow -13(23) + 60(5) = 1$$

$\therefore x = 23, y = 5$ is a particular solution.

General Solution:

$$x = 23 + 60t, y = 5 + 13t, \text{ for any integer } t.$$

$$\text{Verification: } -13(23) + 60(5) = -299 + 300 = 1 \checkmark$$

Q2. What do you mean by the Kerala School of Mathematics? Discuss why this has a significant role in shaping the foundation of modern mathematics in India.

The Kerala School of Mathematics refers to a remarkable lineage of mathematicians and astronomers who flourished in the Kerala region of South India, primarily between the 14th and 16th centuries CE. Founded essentially by Madhava of Sangamagrama (c. 1340–1425 CE), the school produced a series of brilliant scholars including Parameshvara, Nilakantha Somayaji, Jyesthadeva, and Achyuta Pisharati.

Key Significance:

- 1. Pre-Calculus Discoveries:** The Kerala mathematicians independently discovered infinite series expansions for trigonometric functions (sine, cosine, arctangent) and the value of π — achievements conventionally credited to Newton and Leibniz in the 17th century. Madhava's series for $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots$ predates Leibniz by nearly 300 years.
- 2. Rigorous Proof Tradition:** Unlike the sutra-style of earlier Indian mathematics, the Kerala school provided systematic proofs and derivations, documented in Yuktibhasha (c. 1530 CE) — considered one of the first texts in the history of mathematics to contain proofs of series expansions.

3. Astronomy and Mathematics Integration: Their work on infinite series, interpolation, and iterative methods was developed in the context of precise astronomical calculations, giving practical grounding to abstract mathematics.

4. Foundation for Modern Mathematics: The concepts of limits, convergence, and infinite series — cornerstones of modern analysis — were essentially conceptualised by this school, shaping the trajectory of Indian mathematical thought.

Q3. Briefly describe the geographical setting of the Kerala School of Mathematics in the southern part of India.

The Kerala School of Mathematics flourished along the banks of the Nila River (also called the Bharathapuzha), the second longest river in Kerala, in the Malappuram and Palakkad districts of present-day Kerala, India. The region is often called the 'Nila basin' and the school is sometimes referred to as the 'Nila School' of mathematics.

Geographical Highlights:

- The school was centred in the village of **Sangamagrama** (identified with modern Irinjalakuda or Trissur area), birthplace of Madhava.
 - Key associated villages include **Alathiyur** (Alattiyur), **Thrikkandiyur**, and **Kudallur** — all situated in the fertile Nila river valley.
 - The region was under the patronage of the **Zamorin (Samuthiri) kings** of Calicut (Kozhikode), who supported scholarly activities.
 - The tropical, agrarian landscape provided a stable socio-economic base. The Nambudiri Brahmin community, which dominated the scholarly class, maintained a tradition of Vedic learning combined with astronomical observation.
 - The area's relative isolation from the turmoil of North India allowed uninterrupted scholarly transmission across generations.
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Q4. Discuss the key contributions of Madhava of Sangamagrama in the area of algebra and calculus during the 14th century.

Madhava of Sangamagrama (c. 1340–1425 CE) is regarded as the greatest mathematician-astronomer of medieval India and the founder of the Kerala School. His contributions, preserved through later commentaries, represent a quantum leap in mathematical thought.

Contributions in Calculus and Infinite Series:

1. Madhava-Leibniz Series (Gregory–Leibniz series):

$$\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots \text{ (arctan series for } x=1 \text{)}$$

Madhava derived this ~300 years before Leibniz (1673) and Gregory (1671).

2. Sine Series (Madhava–Newton series):

$$\sin \theta = \theta - \theta^3/3! + \theta^5/5! - \dots$$

3. Cosine Series:

$$\cos \theta = 1 - \theta^2/2! + \theta^4/4! - \dots$$

4. Arctangent Series:

$$\arctan x = x - x^3/3 + x^5/5 - \dots \text{ (for } |x| \leq 1 \text{)}$$

5. Highly Accurate Value of π : Madhava computed π correct to 11 decimal places using end-correction terms to accelerate convergence of his series.

6. Interpolation for Trigonometric Tables: He developed second-order interpolation formulas for computing sine values at intermediate angles — a precursor to numerical analysis.

Contributions in Algebra:

- Extended methods for solving indeterminate equations.

- Worked on cyclic quadrilaterals and their properties.
 - Developed systematic methods for approximation and iteration.
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Q5. Who was Nilakantha Somayaji? How did he contribute in mathematics and astronomy through his writings?

Nilakantha Somayaji (1444–1544 CE) was a highly distinguished mathematician, astronomer, and scholar of the Kerala School. Born in the Trikkantiyur village near the Nila river, he was a student of Damodara (son of Parameshvara) and a key transmitter of Madhava's discoveries.

Major Works and Contributions:

1. **Tantrasangraha (1501 CE):** His most celebrated work — a comprehensive treatise on astronomy. It contains revised planetary models, including a partially heliocentric model where the five visible planets revolve around the Sun, which in turn revolves around the Earth. This predates Tycho Brahe's model by decades.
2. **Aryabhatiyabhasya:** An extensive commentary on Aryabhata's *Aryabhatiya*, where Nilakantha provides detailed mathematical proofs and corrections. He gives a rigorous proof of the Madhava sine series.
3. **Golasara and Siddhantadarpana:** Shorter astronomical treatises dealing with spherical astronomy.
4. **Jyotirmimamsa:** Discusses the philosophical basis of astronomical observations and the need for constant revision of parameters.

Mathematical Contributions:

- Improved derivation of the infinite series for π with better convergence estimates.
 - Correct formula for the equation of centre of the Moon's orbit.
 - Discussed the concept of instantaneous velocity (*tatkala-gati*) of planets — a concept akin to differential calculus.
 - Provided geometric proofs for algebraic identities used in astronomy.
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Q6. Discuss the importance of 'Yuktibhasha' for the study of mathematics and astronomy in the Nila School of Mathematics.

Yuktibhasha (meaning 'Rationale in the vernacular language' or 'Language of Proofs') is a seminal text composed by Jyesthadeva (c. 1530 CE), a scholar of the Kerala School. Written in the Malayalam language (making it unusual among Sanskrit mathematical texts), it stands as one of the most remarkable documents in the history of mathematics.

Importance and Significance:

1. **First Systematic Proof Text in Mathematics:** Unlike most Indian mathematical texts that stated results without proof, *Yuktibhasha* provides detailed, step-by-step proofs (*yukti* = rationale/proof) for all major results — including infinite series for π , sine, cosine, and arctangent. This is arguably the first pre-modern proof-based mathematical treatise in the world.
 2. **Preservation of Madhava's Work:** Since Madhava's original works are mostly lost, *Yuktibhasha* is the primary source through which his series expansions and methods are known to modern scholars.
 3. **Covers Both Mathematics and Astronomy:** The text is divided into two halves — the first dealing with pure mathematics (arithmetic, algebra, infinite series, circle geometry) and the second with astronomy (planetary motion, eclipses, spherical astronomy).
 4. **Accessibility:** Being in Malayalam (not Sanskrit), it was accessible to a broader scholarly community and reflects a democratisation of mathematical knowledge.
 5. **Numerical Methods:** Contains sophisticated methods of approximation, iterative computation, and interpolation — techniques fundamental to modern numerical analysis.
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Q7. Briefly elaborate the structural composition of the book 'Yuktibhasha'.

Yuktibhasha is structured into two major parts (Khanda), each further subdivided into chapters (Adhyaya). The text is written in Malayalam prose, making it unique among Indian mathematical literature.

Part I – Mathematics (Ganita Khanda) — Chapters 1–7:

- Ch. 1: **Parikarmaashtakam** — Eight fundamental arithmetic operations, fractions, and their properties.
- Ch. 2: **Dasaprashnangal** — Ten mathematical problems covering basic arithmetic.
- Ch. 3: **Bhinnagaṇitam** — Operations on fractions.
- Ch. 4: **Trairashikam** — Rule of three and its applications.
- Ch. 5: **Kuṭṭakakramam** — The pulveriser method (Kuttaka) for linear Diophantine equations.
- Ch. 6: **Vṛttapariṇāham** — Circumference and diameter of a circle; derivation of the infinite series for π using the arctan series.
- Ch. 7: **Jyānayanam** — Derivation and proof of the infinite series for sine and cosine; construction of sine tables.

Part II – Astronomy (Jyotisha Khanda) — Chapters 8–15:

- Ch. 8–9: Mean and true motions of planets.
- Ch. 10–11: Three problems of astronomical diurnal motion.
- Ch. 12–13: Gnomon problems and shadow calculations.
- Ch. 14–15: Eclipses (lunar and solar), heliacal risings, and the Moon's cusps.

The structure reflects the Kerala School's seamless integration of rigorous mathematics with practical astronomical computation.

Q8. Give an example of a 4×4 magic square whose magic sum is 40.

A magic square of order n has magic sum $S = n(n^2+1)/2$ for consecutive integers starting from 1. For a general magic square with desired sum $S = 40$, we use a construction based on shifting a standard 4×4 magic square.

Standard 4×4 Magic Square (sum = 34):

16	2	3	13
5	11	10	8
9	7	6	12
4	14	15	1

Since $S = 34$ for this standard square, we need to add $(40-34)/4 = 6/4 = 1.5$ to every element. For an integer solution, we use a different base.

Construction for Magic Sum = 40:

Use the formula: start with a base value b such that the new magic sum $= 34 + 4b = 40 \rightarrow b = 1.5$ (non-integer). Alternatively, construct directly using consecutive integers from 7 to 22 (sum of one row = 40):

Sum of 7 to 22 = 16 terms; row sum $= (7+8+\dots+22)/4 = (16 \times 29/2)/4 = 232/4 = 58$. Not 40.

Best approach: Add k to each entry of a standard 4×4 magic square so that the new sum $= 34 + 4k = 40$, giving $k = 1.5$. Since we need integers, we construct as follows — replace $\{1, 2, \dots, 16\}$ with $\{a, a+1, \dots, a+15\}$ where $4a+30 = 40 \rightarrow a = 2.5$. For integer entries, use a direct construction:

4×4 Magic Square with Sum = 40 (using Narayana's method):

25	-3	-1	19
3	15	13	9
11	7	5	17

1	21	23	-5
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Verification: Every row, column, and diagonal sums to 40. ✓

Construction method: Take the standard 4×4 Dürer magic square (sum=34) with entries m_{ij} ; then set new entries = $2 \cdot m_{ij} - 7$. New sum = $2 \times 34 - 4 \times 7 = 40$. ✓

Q9. Find the value of x for which the 3×3 matrix becomes a magic square. Complete the magic square.

Given matrix:

From the PDF, the visible entries are: top-left = $x-1$, top-right = $x-3$, middle = 6, bottom-left = $x+3$, bottom-right = $x+1$.

Step 1: Use the property that the magic sum = $3 \times$ middle element = $3 \times 6 = 18$.

Magic Sum $S = 18$

Step 2: Find x using the main diagonal (top-left to bottom-right):

$$(x-1) + 6 + (x+1) = 18$$

$$2x + 6 = 18$$

$$2x = 12 \rightarrow x = 6$$

Step 3: Substitute $x = 6$ to find all entries:

Top-left: $x-1 = 5$, Top-right: $x-3 = 3$

Bottom-left: $x+3 = 9$, Bottom-right: $x+1 = 7$

Centre = 6 (given)

Step 4: Find remaining entries using magic sum = 18:

Top-middle: $18 - 5 - 3 = 10$ (check: not right for standard 3×3)

Let positions be: a b c / d e f / g h i, with $e=6$, $a=5$, $c=3$, $g=9$, $i=7$

$$\text{Row1: } 5+b+3=18 \rightarrow b=10$$

$$\text{Col2: } 10+6+h=18 \rightarrow h=2$$

$$\text{Row3: } 9+2+7=18 \checkmark$$

$$\text{Col1: } 5+d+9=18 \rightarrow d=4$$

$$\text{Col3: } 3+f+7=18 \rightarrow f=8$$

$$\text{Row2: } 4+6+8=18 \checkmark$$

$$\text{Diag (anti): } 3+6+9=18 \checkmark \quad \text{Diag (main): } 5+6+7=18 \checkmark$$

Completed Magic Square ($x = 6$, $S = 18$):

5	10	3
4	6	8
9	2	7

All rows, columns, and diagonals sum to 18. ✓

Q10. First order interpolating approximations of $\sin(60^\circ+\epsilon)$ and $\cos(90^\circ+\epsilon)$ for $\epsilon = 10^{-2}$ up to 4 digits after decimal.

Madhava (and later Brahmagupta) developed interpolation formulas for trigonometric functions. The first-order (linear) interpolation corresponds to the standard differential approximation.

Derivation — First Order Interpolation Formula:

If $f(\theta)$ is known at θ , the first-order approximation at $\theta + \epsilon$ is:

$$f(\theta + \epsilon) \approx f(\theta) + \epsilon \cdot f'(\theta)$$

Computation for $\sin(60^\circ + \varepsilon)$, $\varepsilon = 10^{-2}$ radians:

$$f(\theta) = \sin \theta, f'(\theta) = \cos \theta$$

$$\theta_0 = 60^\circ = \pi/3$$

$$\sin(60^\circ) = \sqrt{3}/2 = 0.8660$$

$$\cos(60^\circ) = 1/2 = 0.5000$$

$$\sin(60^\circ + \varepsilon) \approx \sin(60^\circ) + \varepsilon \cdot \cos(60^\circ)$$

$$\approx 0.8660 + (0.01)(0.5000)$$

$$\approx 0.8660 + 0.0050$$

$$= 0.8710$$

Computation for $\cos(90^\circ + \varepsilon)$, $\varepsilon = 10^{-2}$ radians:

$$f(\theta) = \cos \theta, f'(\theta) = -\sin \theta$$

$$\theta_0 = 90^\circ = \pi/2$$

$$\cos(90^\circ) = 0$$

$$\sin(90^\circ) = 1$$

$$\cos(90^\circ + \varepsilon) \approx \cos(90^\circ) + \varepsilon \cdot (-\sin(90^\circ))$$

$$\approx 0 + (0.01)(-1)$$

$$= -0.0100$$

Results (4 decimal places):

$$\sin(60^\circ + 0.01) \approx 0.8710$$

$$\cos(90^\circ + 0.01) \approx -0.0100$$

Exact values: $\sin(60^\circ + 0.01) = 0.8760$ (to 4dp) — the approximation is accurate to first order.

Q11. Second order interpolating approximations of $\sin(0^\circ + \varepsilon)$ and $\cos(45^\circ + \varepsilon)$ for $\varepsilon = 10^{-4}$ up to 6 digits after decimal.

Second Order (Madhava) Interpolation Formula:

$$f(\theta_0 + \varepsilon) \approx f(\theta_0) + \varepsilon \cdot f'(\theta_0) + (\varepsilon^2/2) \cdot f''(\theta_0)$$

For $\sin(0^\circ + \varepsilon)$, $\varepsilon = 10^{-4}$:

$$f(\theta) = \sin \theta, f'(\theta) = \cos \theta, f''(\theta) = -\sin \theta$$

$$\text{At } \theta_0 = 0^\circ: \sin(0^\circ) = 0, \cos(0^\circ) = 1, -\sin(0^\circ) = 0$$

$$\sin(0^\circ + \varepsilon) \approx 0 + \varepsilon \cdot (1) + (\varepsilon^2/2) \cdot (0)$$

$$= \varepsilon = 10^{-4} = 0.000010$$

$$\text{Result: } \sin(0^\circ + 10^{-4}) \approx 0.000010$$

For $\cos(45^\circ + \varepsilon)$, $\varepsilon = 10^{-4}$:

$$f(\theta) = \cos \theta, f'(\theta) = -\sin \theta, f''(\theta) = -\cos \theta$$

$$\text{At } \theta_0 = 45^\circ = \pi/4:$$

$$\cos(45^\circ) = 1/\sqrt{2} = 0.707107$$

$$-\sin(45^\circ) = -1/\sqrt{2} = -0.707107$$

$$-\cos(45^\circ) = -1/\sqrt{2} = -0.707107$$

$$\cos(45^\circ + \varepsilon) \approx 0.707107 + (10^{-4})(-0.707107) + ((10^{-4})^2/2)(-0.707107)$$

$$\approx 0.707107 - 0.000007071 - 0.000000000000035$$

$$\approx 0.707107 - 0.000007 \text{ (higher terms negligible)}$$

$$\approx 0.707100$$

Results (6 decimal places):

$$\sin(0^\circ + 10''') \approx 0.000010$$

$$\cos(45^\circ + 10''') \approx 0.707100$$

Q12. Give an account of the infinite series representations of several functions proposed by Madhava of Sangamagrama.

Madhava of Sangamagrama (c. 1340–1425 CE) was the first mathematician in history to systematically develop infinite series expansions for trigonometric functions and for π . His results, preserved in the works of his successors (especially Yuktibhasha), include:

1. Madhava–Leibniz Series for π (Arctangent Series at $x=1$):

$$\pi/4 = 1 - 1/3 + 1/5 - 1/7 + 1/9 - \dots$$

$$(\arctan x = x - x^3/3 + x^5/5 - x^7/7 + \dots, \text{ for } |x| \leq 1)$$

2. Sine Series:

$$\sin \theta = \theta - \theta^3/3! + \theta^5/5! - \theta^7/7! + \dots$$

3. Cosine Series:

$$\cos \theta = 1 - \theta^2/2! + \theta^4/4! - \theta^6/6! + \dots$$

4. Arctangent Series (General):

$$\arctan x = x - x^3/3 + x^5/5 - x^7/7 + \dots, |x| \leq 1$$

5. Improved π Series with End-Correction Terms:

Madhava improved convergence of the $\pi/4$ series by adding rational correction terms. He computed π to 11 decimal places as:

$$\pi \approx 3.14159265359$$

6. Madhava's value in verse (from Mahajyanayana):

$$\pi \approx 2827433388233/900000000000 \approx 3.14159265359 \text{ (correct to 11 decimal places)}$$

These discoveries predate Newton's general binomial series (1665) and Taylor's theorem (1715) by approximately 250–300 years, establishing Madhava as a founder of mathematical analysis.

Q13. What is the Madhava-Leibnitz series? Discuss the convergence of the series.

The Madhava-Leibniz Series:

$$\pi/4 = 1 - 1/3 + 1/5 - 1/7 + 1/9 - \dots = \sum_{n=0}^{\infty} (-1)^n / (2n+1)$$

This series was discovered by Madhava of Sangamagrama (~1400 CE), and independently rediscovered by James Gregory (1671) and Gottfried Wilhelm Leibniz (1673). It follows from the arctangent series:

$$\arctan x = x - x^3/3 + x^5/5 - x^7/7 + \dots \text{ (for } |x| \leq 1)$$

Setting $x = 1$ gives $\arctan(1) = \pi/4$, which yields the series above.

Convergence Analysis:

- The series is an **alternating series** of the form $\sum (-1)^n a_n$ where $a_n = 1/(2n+1)$.
- By the **Alternating Series Test (Leibniz criterion)**:
 - (i) $a_n = 1/(2n+1) > 0$ for all n ✓
 - (ii) a_n is decreasing: $1/(2n+1) > 1/(2n+3)$ ✓
 - (iii) $\lim_{n \rightarrow \infty} a_n = \lim 1/(2n+1) = 0$ ✓
- All three conditions are satisfied, so the series **converges** (conditionally).
- The series does NOT converge absolutely since $\sum 1/(2n+1)$ diverges (harmonic-type).
- **Rate of convergence** is very slow — to get 10 correct decimal places of π requires millions of terms. Madhava overcame this by adding correction terms, dramatically accelerating convergence.

Error Bound (from Alternating Series Estimate):

$$|\text{Error}| \leq a_{n+1} = 1/(2n+3)$$

Madhava's correction terms effectively transformed this into a rapidly convergent series, allowing him to compute π to 11 decimal places.

Q14. Find the solution of quadratic Diophantine equation $Nx^2 + C = y^2$ for $N=2$, $C=1$.

The equation is: $2x^2 + 1 = y^2$

Method 1: Trial and Error

x	$2x^2+1$	$y^2?$	Integer y?
1	3	—	No
2	9	9	Yes, y=3
5	51	—	No
12	289	289	Yes, y=17

Method 2: Bhavana (Composition Law) by Brahmagupta

The Bhavana principle states: If (x_1, y_1) satisfies $Nx^2 + C = y^2$ and (x_2, y_2) satisfies $Nx^2 + C = y^2$, then $(x_1y_2 + x_2y_1, y_1y_2 + Nx_1x_2)$ satisfies $Nx^2 + C = y^2$.

Trivial solution: $x=1, y=1$: $2(1)^2+1=3 \neq 1^2$. Try $x=0, y=1$: $2(0)+1=1=1^2$. ✓

Trivial: $(x_1, y_1) = (0, 1)$, $C_1 = 1$

Start with: $(x, y) = (1, 1)$: $N \cdot 1 + C = 2 + C = y^2 \rightarrow$ need $y^2=3$ (not perfect square)

Try: seed $(1, 1)$ with $2 \cdot 1^2 + (-1) = 1^2 \rightarrow 2x^2 - 1 = y^2$. So $(1, 1)$ satisfies $2x^2 - 1 = y^2$.

Apply Bhavana: compose $(1, 1)$ with itself ($C=-1$):

$x' = 1 \cdot 1 + 1 \cdot 1 = 2$, $y' = 1 \cdot 1 + 2 \cdot 1 \cdot 1 = 3$, $C' = (-1)(-1) = 1$

Verify: $2(2^2)+1 = 8+1 = 9 = 3^2$ ✓

Apply again: compose $(2, 3)$ with $(1, 1)$, $C_1=1$, $C_2=-1$:

$x' = 2 \cdot 1 + 1 \cdot 3 = 5$, $y' = 3 \cdot 1 + 2 \cdot 2 \cdot 1 = 7$, $C' = 1 \cdot (-1) = -1$

$2(5^2)-1=50-1=49=7^2$ ✓ $\rightarrow (5, 7)$ satisfies $2x^2-1=y^2$

Compose $(5, 7)$ with itself: $x'=5 \cdot 7 + 7 \cdot 5 = 70$... $\rightarrow (70, 99)$ for $C=1$

Check: $2(70)^2+1 = 9800+1=9801=99^2$ ✓

Fundamental solutions for $2x^2+1=y^2$: $(x, y) = (2, 3), (12, 17), (70, 99), \dots$

Q15. Find the solution of Pell's equation $Nx^2 + C = y^2$ for $N=3$, $C=1$ using Bhavana and Chakravala.

Pell's equation: $x^2 \cdot 3 + 1 = y^2$, i.e., $3x^2 + 1 = y^2$ (find positive integer solutions).

Step 1: Seed solution using trial:

$x=0, y=1$: $3(0)+1=1=1^2$ ✓ (trivial)

$x=1, y=2$: $3(1)+1=4=2^2$ ✓ but $C=1$: $3 \cdot 1 + 1 = 4 = 2^2$ ✓

Wait: $3(1^2)+1=4=2^2 \rightarrow (x, y)=(1, 2)$ is a solution! ✓

Step 2: Chakravala (Cyclic Method) to verify:

Chakravala: find m such that $|m^2-N|$ is minimised and $m \equiv y/x \pmod{k}$

$N=3$: $\sqrt{3} \approx 1.732$. Start with $m=2$ (nearest integer).

$k = m^2 - N = 4 - 3 = 1 \rightarrow k=1$ means we already have a solution!

So $(x, y)=(1, 2)$ with $C=1$ is indeed the fundamental solution.

Step 3: Generate more solutions using Bhavana:

Compose $(1, 2)$ with itself ($C_1=C_2=1$):

$x' = x_1y_2 + x_2y_1 = 1 \cdot 2 + 1 \cdot 2 = 4$

$y' = y_1y_2 + N \cdot x_1x_2 = 2 \cdot 2 + 3 \cdot 1 \cdot 1 = 7$

$$C' = C \cdot C = 1 \cdot 1 = 1$$

$$\text{Check: } 3(4^2)+1 = 48+1 = 49 = 7^2 \checkmark$$

Next: compose (4,7) with (1,2):

$$x' = 4 \cdot 2 + 1 \cdot 7 = 15, y' = 7 \cdot 2 + 3 \cdot 4 = 26$$

$$3(15^2)+1=675+1=676=26^2 \checkmark$$

Solutions: $(x,y) = (1,2), (4,7), (15,26), (56,97), \dots$

Q16. Solve $92x^2 + 1 = y^2$ using Bhavana to transform fractional solutions to integer solutions.

We need integer solutions to the Pell-type equation: $92x^2 + 1 = y^2$.

Step 1: Find a near-solution (fractional seed):

$$\sqrt{92} \approx 9.591. \text{ Try } y/x \approx 9.591.$$

$$\text{For } x=1: 92+1=93 \text{ (not perfect square)}$$

$$\text{For } x=2: 92 \cdot 4 + 1 = 369 \text{ (not perfect square)}$$

$$\text{For } x=5: 92 \cdot 25 + 1 = 2301 \text{ (not perfect square)}$$

$$\text{For } x=10: 92 \cdot 100 + 1 = 9201 \text{ (not perfect square; } \sqrt{9201} \approx 95.9)$$

$$\text{For } x=120: 92 \cdot 14400 + 1 = 1324801 = 1151^2 \checkmark$$

Using Chakravala algorithm:

$$\sqrt{92} \approx 9.591. p=9, k=9^2-92=81-92=-11$$

m : find m such that $p+m$ divisible by $|k|=11$ and m closest to $\sqrt{92}$

$$m=(-9+11 \cdot t) \text{ for integer } t; m=2: \text{ check } |m^2-92|=|4-92|=88; k=(4-92)/-11=8$$

$$m=9: (81-92)/-11=1; \text{ this gives } k=1 \text{ — solution found quickly!}$$

Chakravala solution for $92x^2 + 1 = y^2$:

$$p=9, k=-11; m=2 \rightarrow k=(4-92)/(-11)=8$$

$$p=2+(11 \cdot 0)=2 \dots \text{ (continue iterations)}$$

After full Chakravala: fundamental solution is $(x,y) = (120, 1151)$

$$\text{Verify: } 92 \times 120^2 + 1 = 92 \times 14400 + 1 = 1324800 + 1 = 1324801 = 1151^2 \checkmark$$

Fundamental Solution: $x = 120, y = 1151$

Q17. Solve $3x^2 - 800 = y^2$ using Chakravala rational solutions and scaling method.

Equation: $3x^2 - 800 = y^2$, i.e., $3x^2 - y^2 = 800$.

Step 1: Find a rational/approximate solution:

$$\text{Rearrange: } y^2 = 3x^2 - 800$$

$$\text{For } y \text{ real: } 3x^2 > 800 \rightarrow x > \sqrt{(800/3)} \approx 16.33, \text{ so } x \geq 17.$$

$$x=17: 3 \cdot 289 - 800 = 867 - 800 = 67 \text{ (not square)}$$

$$x=20: 3 \cdot 400 - 800 = 400 = 20^2 \checkmark \rightarrow (x,y)=(20,20) \text{ is a solution!}$$

Verify: $3(20^2) - 800 = 1200 - 800 = 400 = 20^2 \checkmark$

So $(x, y) = (20, 20)$ is a valid integer solution.

Step 2: Generate more solutions using Bhavana/composition:

The equation $3x^2 - y^2 = 800$ can be written as $3x^2 + (-1)y^2 = 800$. We use the identity for Pell-like equations.

From the auxiliary equation $3a^2 - b^2 = 1$ (Pell's equation for $N=3$):

$$\text{Solutions: } (a,b) = (1,\sqrt{2}) \dots \text{ better: use } 3a^2 - b^2 = -1: (a,b)=(1,2) \text{ since } 3-4=-1 \checkmark$$

Composition of $(x,y)=(20,20)$ with $(a,b)=(1,2)$ for $C=-1$:

$$\text{New solution: } x' = x \cdot b + y \cdot a \cdot (N) / \dots \text{ (generalised Bhavana)}$$

$$x' = 20 \cdot 2 + 20 \cdot 1 = 60, y' = 20 \cdot 2 + 3 \cdot 20 \cdot 1 = 100$$

$$\text{Check: } 3(60^2) - (100^2) = 10800 - 10000 = 800 \checkmark$$

Solutions: (x,y) = (20,20), (60,100), and infinitely many more via composition.

Q18. Why is Brahmagupta's formula considered a generalization of Heron's formula?

Heron's Formula (for a triangle with sides a, b, c):

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}, \text{ where } s = (a+b+c)/2$$

Brahmagupta's Formula (for a cyclic quadrilateral with sides a, b, c, d):

$$\text{Area} = \sqrt{[(s-a)(s-b)(s-c)(s-d)]}, \text{ where } s = (a+b+c+d)/2$$

Why it is a Generalization:

1. **Triangle as degenerate quadrilateral:** A triangle can be viewed as a cyclic quadrilateral where one side has length $d = 0$. Setting $d = 0$ in Brahmagupta's formula:

$$s = (a+b+c+0)/2 = (a+b+c)/2 \text{ (same as Heron's } s)$$

$$\text{Area} = \sqrt{[(s-a)(s-b)(s-c)(s-0)]} = \sqrt{s(s-a)(s-b)(s-c)}$$

This is exactly **Heron's formula**. ✓

2. **Scope:** Heron's formula works only for triangles; Brahmagupta's works for any cyclic quadrilateral (a quadrilateral inscribed in a circle).

3. **Historical context:** Brahmagupta (628 CE) discovered his formula as a natural extension of Heron's result (~60 BCE), unifying the area computation for polygons inscribed in circles.

4. **Note:** Brahmagupta's formula gives the maximum area for a quadrilateral with given sides — achieved when the quadrilateral is cyclic. For non-cyclic quadrilaterals, the formula is an overestimate.

Q19. Prove Narayana's beautiful area theorem: $2AD = d_1 d_2 d_3$. Discuss reductions and limitations.

Narayana's Area Theorem:

For a cyclic quadrilateral ABCD inscribed in a circle, with the three diagonals AC (= d_1), BD (= d_2), and the 'third diagonal' EF (= d_3) where E, F are the intersections of extensions of opposite sides (or the diagonal of the complete quadrilateral), and area = AD:

$$2 \cdot \text{Area}(ABCD) = d_1 \cdot d_2 \cdot d_3 / R \text{ (in some formulations)}$$

In Narayana Pandita's formulation (Ganitakaumudi, 14th c.), for a cyclic quadrilateral:

$$2 \cdot AD = d_1 \cdot d_2 \cdot d_3$$

where the three diagonals are defined using the complete quadrilateral construction.

Proof Sketch:

1. Use Ptolemy's theorem: $d_1 \cdot d_2 = ac + bd$ (product of diagonals = sum of products of opposite sides).
2. The third diagonal d_3 relates the area via: $\text{Area} = (d_1 \cdot d_2 \cdot \sin \theta) / 2$ where θ is the angle between diagonals.
3. Using the extended law of sines and properties of cyclic quadrilaterals, the relationship $2 \cdot \text{Area} = d_1 \cdot d_2 \cdot d_3 / (2R)$ can be established, where R is the circumradius.

When one diagonal becomes the diameter ($d_1 = 2R$):

The inscribed angle theorem gives: all angles subtended by the diameter = 90° .

The quadrilateral degenerates toward Brahmagupta's configuration.

Brahmagupta's formula is recovered: $\text{Area} = \sqrt{[(s-a)(s-b)(s-c)(s-d)]}$.

Why Brahmagupta's formula fails for non-cyclic quadrilaterals:

- Brahmagupta's formula assumes the quadrilateral is inscribed in a circle (cyclic).

- For a non-cyclic quadrilateral, there is no unique circumscribed circle, and the area depends on the angles as well as the sides.
- The generalisation to non-cyclic quadrilaterals requires the Bretschneider formula: $\text{Area} = \sqrt{[(s-a)(s-b)(s-c)(s-d) - abcd \cdot \cos^2((A+C)/2)]}$.

Q20. Explain the structural difference between Greek and Indian cyclic geometry. Derive Ptolemy's theorem from Narayana's theorem.

Structural Differences:

Aspect	Greek Geometry	Indian Geometry
Approach	Axiomatic-deductive (Euclid's Elements)	Computational-constructive (Sulbasutras, Ganitakaumudi)
Focus	Logical proof from axioms	Practical results with proof (yukti)
Cyclic Quadrilateral	Ptolemy's theorem: $d \cdot d = ac + bd$	Brahmagupta's area formula; Narayana's three-diagonal theorem
Diagonal Concept	Two diagonals only	Third diagonal of complete quadrilateral introduced
Area Formula	Derived geometrically	Algebraic formula for area with proof (Yuktibhasha)

Deriving Ptolemy's Theorem from Narayana's Framework:

For a cyclic quadrilateral ABCD with sides $a=AB$, $b=BC$, $c=CD$, $d=DA$ and diagonals $p=AC$, $q=BD$:

Narayana's theorem (using the area relationship and the third diagonal $r=EF$):

$$\text{Area}(ABCD) = (p \cdot q \cdot r)/(4R)$$

Ptolemy's theorem states: $p \cdot q = a \cdot c + b \cdot d$.

This follows from the Narayana framework by considering the ratio of the area expressed via the two triangulations of the quadrilateral and applying the extended sine rule. Specifically, decompose ABCD into triangles ABC and ACD:

$$\text{Area} = (1/2) \cdot p \cdot q \cdot \sin \theta \text{ (where } \theta = \text{angle between diagonals)}$$

Using the sine rule in each triangle and the cyclic property, one derives $p \cdot q = ac + bd$ — Ptolemy's theorem.

Q21. Show analytically why a cyclic quadrilateral has only three independent sides. Show how area symmetry appears.

Three Independent Sides:

A cyclic quadrilateral ABCD inscribed in a circle of radius R has four sides a, b, c, d and four angles A, B, C, D. The constraint that all four vertices lie on a circle imposes the relation:

$$A + C = \pi \text{ and } B + D = \pi \text{ (opposite angles are supplementary)}$$

This means knowing three sides and one angle (or equivalently three sides and the radius R) completely determines the quadrilateral. The fourth side is determined by the constraint that the quadrilateral must close — i.e., the sum of the arcs must equal 2π :

$$\text{Using the extended sine rule: } a = 2R \sin(\alpha), b = 2R \sin(\beta), c = 2R \sin(\gamma), d = 2R \sin(\delta)$$

$$\text{where } \alpha + \beta + \gamma + \delta = \pi. \text{ Given } \alpha, \beta, \gamma \rightarrow \delta = \pi - \alpha - \beta - \gamma \text{ is determined.}$$

Therefore, only three of the four sides are independent; the fourth is determined by the cyclic constraint.

Area Symmetry in Cyclic Quadrilaterals:

The area of a cyclic quadrilateral (by Brahmagupta's formula) is:

$$K = \sqrt{[(s-a)(s-b)(s-c)(s-d)]}$$

This expression is symmetric in a, b, c, d — swapping any two sides gives the same area. This is the area symmetry: any permutation of the sides of a cyclic quadrilateral gives the same area (as long as the quadrilateral remains cyclic and convex). This is because the circumradius and hence the total area depend only on the set $\{a, b, c, d\}$, not their arrangement.

Q22. Explain mathematically how altitude separation measures deviation from orthogonality. Why does cyclic quadrilateral theory represent interaction between algebra and geometry?

Altitude Separation and Deviation from Orthogonality:

In a cyclic quadrilateral $ABCD$, let the diagonals AC and BD intersect at point P . Define the altitude separation as the perpendicular distances from P to the sides of the quadrilateral. If the diagonals are orthogonal (perpendicular), the altitude separation is zero (symmetric). If not, the difference measures the deviation.

For diagonals AC and BD intersecting at angle θ :

$$\text{Area} = (1/2) \cdot AC \cdot BD \cdot \sin \theta$$

When $\theta = 90^\circ$, $\sin \theta = 1$, giving maximum area for given diagonal lengths. Any deviation from $\theta = 90^\circ$ reduces the area. The 'altitude separation' can be quantified by the angle $(90^\circ - \theta)$, which measures non-orthogonality.

$$\text{Deviation from orthogonality} = |90^\circ - \theta| = |\arccos(d_{AC} \cdot d_{BD} / (|d_{AC}| |d_{BD}|))|$$

Algebra-Geometry Interaction in Cyclic Quadrilateral Theory:

- **Ptolemy's theorem** ($p \cdot q = ac + bd$) is an algebraic identity encoding geometric structure.
- **Brahmagupta's area formula** translates geometric configuration into a symmetric polynomial.
- **Bhavana (composition law)** for Diophantine equations parallels the composition of geometric configurations.
- The Kerala school's use of infinite series to compute circumradii and areas shows how algebraic tools (series, limits) solve geometric problems.
- This integration — algebra providing tools, geometry providing meaning — is the hallmark of the Indian mathematical tradition and anticipates modern algebraic geometry.

Q23. Explain the significance of Narayana's third diagonal. Why was it revolutionary mathematically?

Narayana Pandita (14th century CE), in his *Ganitakaumudi*, introduced the concept of the 'third diagonal' of a cyclic quadrilateral — a concept absent in earlier Indian and Greek treatments.

What is the Third Diagonal?

For a cyclic quadrilateral $ABCD$, the standard two diagonals are AC and BD . Narayana extended the sides AB, DC to meet at a point E , and the sides AD, BC to meet at a point F . The line segment EF is the third diagonal. Together, AC, BD , and EF form the complete quadrilateral associated with $ABCD$.

Mathematical Significance:

1. **Enriched Metric Structure:** The three diagonals satisfy a beautiful multiplicative area theorem ($2 \cdot \text{Area} = d_{AC} \cdot d_{BD} \cdot d_{EF} / \text{constant}$), relating all three lengths to the area in one elegant formula.
2. **Symmetry:** The complete quadrilateral has three pairs of opposite vertices (three 'diagonals'), revealing a hidden symmetry not visible with just two diagonals.
3. **Anticipates Projective Geometry:** The complete quadrilateral is a fundamental concept in projective geometry (developed in Europe centuries later by Desargues and Pascal). Narayana's introduction of the third diagonal thus anticipates projective ideas.
4. **Generalisation of Ptolemy:** With the third diagonal, Narayana's theorem generalises Ptolemy's theorem from a product identity to an area identity involving all three diagonals.

5. **Revolutionary Impact:** It shifted the focus from just the quadrilateral to its associated complete figure, opening a richer geometric landscape — a conceptual leap that is the hallmark of genuine mathematical innovation.

Q24. Using Vedic Sutras solve the given problems.

a) 'Yaavadunam' (Sutra: 'By how much less / more than the base'): Used for squaring numbers near powers of 10.

Principle: For n near base B : $n^2 = [(n-d) \cdot B + d^2]$ where $d = n - B$. For numbers near 100: $n^2 = (n + (n-100)) \times 100 + (100-n)^2$

$$969^2 : \text{base}=1000, d=-31. 969^2=(969-31) \times 1000 + 31^2=938 \times 1000 + 961=938961$$

$$92^2 : \text{base}=100, d=-8. 92^2=(92-8) \times 100 + 8^2=84 \times 100 + 64=8464$$

$$104^2 : \text{base}=100, d=+4. 104^2=(104+4) \times 100 + 4^2=108 \times 100 + 16=10816$$

$$993^2 : \text{base}=1000, d=-7. 993^2=(993-7) \times 1000 + 7^2=986 \times 1000 + 49=986049$$

b) 'Ekanyunena Purvena' (Sutra: 'By one less than the previous'): For multiplying by series of 9s.

Principle: $A \times (10^{\blacksquare} - 1) = A \times 10^{\blacksquare} - A$. Result has two parts: $(A-1)$ and (complement of A to $10^{\blacksquare}$).

$$9999 \times 2378 = 23780000 - 2378 = 23777622$$

$$347 \times 99999 = 34700000 - 347 = 34699653$$

$$999 \times 324567 = 324567000 - 324567 = 324242433$$

$$8765 \times 9999 = 87650000 - 8765 = 87641235$$

c) 'Anurupyena' (Sutra: 'Proportionately'): For numbers of the form $(m+a)(m+b)=m^2+(a+b)m+ab$.

Principle: $(50+a)(50-a)=2500-a^2$, or use $(a+b)(c+d)=ac+ad+bc+bd$ structured.

$$48 \times 42 = (45+3)(45-3) = 45^2 - 9 = 2025 - 9 = 2016$$

$$46 \times 44 = (45+1)(45-1) = 45^2 - 1 = 2025 - 1 = 2024$$

$$61 \times 43 = (52+9)(52-9) = 52^2 - 81 = 2704 - 81 = 2623$$

$$47 \times 43 = (45+2)(45-2) = 45^2 - 4 = 2025 - 4 = 2021$$

d) 'Dhvajanka' (Sutra: 'On top of the flag'): Vedic long division method.

Principle: The divisor is split; the last digit becomes the 'flag' digit used for recursive correction.

$$38982 \div 73 = 534 \text{ (verify: } 73 \times 534 = 38982 \text{ ✓)}$$

$$7332 \div 64 = 114 \text{ r } 36 \text{ (} 64 \times 114 = 7296, 7332 - 7296 = 36 \text{)}$$

$$601325 \div 76 = 7912 \text{ r } 13 \text{ (} 76 \times 7912 = 601312, 601325 - 601312 = 13 \text{)}$$

$$37941 \div 47 = 807 \text{ r } 12 \text{ (} 47 \times 807 = 37929, 37941 - 37929 = 12 \text{)}$$

e) 'Paraavartya Yojayet' (Sutra: 'Transpose and adjust'): Division by algebraic manipulation.

Principle: Divide by inverting the divisor's non-leading digits and adding (transposing and adjusting remainders).

$$289487 \div 13103 \approx 22 \text{ r } 1241 \text{ (} 13103 \times 22 = 288266, \text{ remainder} = 1221 \text{)}$$

$$497342 \div 121 = 4110 \text{ r } 12 \text{ (} 121 \times 4110 = 497310, \text{ remainder} = 32 \text{)} \approx 4110 \text{ r } 32$$

$$36520981 \div 133 = 274594 \text{ r } 39 \text{ (} 133 \times 274594 = 36521002 \dots \text{ let's compute: } 133 \times 274594 = 36521002 \text{:}$$

$$\text{adjust} \rightarrow 274593 \text{ r } 120 \text{)}$$

$$136972 \div 121 = 1132 \text{ r } 0 \text{ (} 121 \times 1132 = 136972 \text{ ✓ exact)}$$

f) 'Urdhva-Tiryagbhyam' (Sutra: 'Vertically and crosswise'): General multiplication.

Principle: Multiply digit by digit vertically and crosswise, carrying as needed.

$$94 \times 78:$$

$$\text{Step1 (units): } 4 \times 8 = 32, \text{ write 2 carry 3}$$

$$\text{Step2 (cross): } 9 \times 8 + 4 \times 7 = 72 + 28 = 100, +3 = 103, \text{ write 3 carry 10}$$

$$\text{Step3 (hundreds): } 9 \times 7 = 63, +10 = 73$$

$$\text{Result: } 7332$$

103×105 :

Cross: $103 \times 105 = 100 \times 105 + 3 \times 105 = 10500 + 315 = 10815$

By Anurupyena: $(100+3)(100+5) = 10000 + 800 + 15 = 10815$ ✓

24×36 :

Step1: $4 \times 6 = 24$, write 4 carry 2

Step2: $2 \times 6 + 4 \times 3 = 12 + 12 = 24$, $+2 = 26$, write 6 carry 2

Step3: $2 \times 3 = 6$, $+2 = 8$

Result: 864

252×846 :

$= 200 \times 846 + 52 \times 846 = 169200 + 43992 = 213192$

— End of Assignment 3 Answer Key —
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